

## Output-Procedure Call Diagnostic test

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260405696

Lab Time 16:00 Desk Number: Desk- G-084

### How to work out what a code fragment does: draw a table

The simplest and most reliable way to work out what a code fragment does (and to understand it) is to draw a table. This is called tracing or dry running code. Let's work out what this fragment does:

```
System.out.print("Holding out for");           // L1
System.out.println("a hero");                   // L2
System.out.println("is by Bonnie Tyler.");       // L3
```

#### Executing Line L1

This prints out the given string. Important to notice is it uses `System.out.print` not `System.out.println`. This means anything that follows goes on the same line with no spaces unless there is one in the String (there isn't here)

| LINE | printed         |
|------|-----------------|
| L1   | Holding out for |

#### Executing Line L2

This prints on the same line as the previous text printed but at the end goes onto a new line so that anything else printed in later lines will go to a new line. No spaces are printed after "for".

| LINE | printed         |
|------|-----------------|
| L1   | Holding out for |
| L2   | a hero          |

#### Executing Line L3

This now prints the new string on the new line. As it uses `println` the cursor finishes at the start of the next line.

| LINE | printed             |
|------|---------------------|
| L1   | Holding out for     |
| L2   | a hero              |
| L3   | is by Bonnie Tyler. |

#### Overall what is printed is:

```
Holding out for
a hero
is by Bonnie Tyler
```

Notice how the title of the song has been printed on one line and the artist on the next, but that a space is missing - we have found a bug! We need to add an explicit space in the Line L1 String.

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You must show all your working in the following. No correct table (exactly the same as the solution)... no marks. If you do not get them all right and do not understand why, ask for help.

**Complete the dry run table** for each fragment of code given in the questions below. **Give** the value(s) printed to the screen when each is executed.

**Follow the module dry run style guide in your answers.**

1.

```
System.out.print("I am ");           // L1
System.out.println("the one and only"); // L2
```

**Complete the dry run table (you may not need all rows):**

| LINE | printed          |
|------|------------------|
| L1   | I am             |
| L2   | the one and only |
|      |                  |

**Write EXACTLY what is printed here**

I am the one and only

2.

```
System.out.println("Sisters are doin' it"); // L1
System.out.println("for themselves");       // L2
```

**Complete the dry run table (you may not need all rows):**

| LINE | printed              |
|------|----------------------|
| L1   | Sisters are doin' it |
| L2   | for themselves       |
|      |                      |

**Write EXACTLY what is printed here**

Sisters are doin' it  
for themselves

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3.

```
System.out.print("I will survive ");           // L1
System.out.println("by");                       // L2
System.out.println("Gloria Gaynor");           // L3
```

Complete the dry run table (you may not need all rows):

| LINE | printed        |
|------|----------------|
| L1   | I will survive |
| L2   | by             |
| L3   | Gloria Gaynor  |

Write EXACTLY what is printed here

I will survive by  
Gloria Gaynor

4.

```
System.out.println("I will survive");           // L1
System.out.print("by");                         // L2
System.out.println("Gloria Gaynor");           // L3
```

Complete the dry run table (you may not need all rows):

| LINE | printed        |
|------|----------------|
| L1   | I will survive |
| L2   | by             |
| L3   | Gloria Gaynor  |

Write EXACTLY what is printed here

I will surviveby  
Gloria Gaynor

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### How to work out what a procedure call does: draw more tables

With procedures you draw a new table for each procedure call. Consider:

```
public static void printBornToRun ()
{
    System.out.print("Born to Run");           // L1
    return;
}
public static void printSong ()
{
    System.out.print("Springsteen wrote ");    // L2
    printBornToRun();                          // L3
    return;
}
```

with a test call      `printSong ();`

#### Executing Line L2

Notice here we execute line L2 first as we call the method `printSong` which just prints a string.

| printSong() |                   |
|-------------|-------------------|
| LINE        | printed           |
| L2          | Springsteen wrote |

#### Executing Line L3

We have to pause filling out the table and start a new table for `printBornToRun`. Once it has finished executing we will be able to return to complete the original table.

| printSong() |                   |
|-------------|-------------------|
| LINE        | printed           |
| L2          | Springsteen wrote |
| L3          |                   |

| printBornToRun() |             |
|------------------|-------------|
| LINE             | printed     |
|                  | Born to Run |

#### Executing Line L1

As we are now executing the procedure we execute Line L1. which prints Born to Run then returns back to where we came from so Line L2.

| printBornToRun() |             |
|------------------|-------------|
| LINE             | printed     |
| L1               | Born to run |

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### Continuing Executing Line L3

As we are now executing the original procedure again so return to the printSong table. The previous table has shown us that the call to printBornToRun printed out Born to Run so we note that that is what Line L3 prints

| printSong() |                   |
|-------------|-------------------|
| LINE        | printed           |
| L3          | Springsteen wrote |
| L4          | Born to run       |

We therefore end up with the final two tables:

| printSong() |                   |
|-------------|-------------------|
| LINE        | printed           |
| L3          | Springsteen wrote |
| L4          | Born to run       |

| printBornToRun() |             |
|------------------|-------------|
| LINE             | printed     |
| L1               | Born to run |

### Overall what is printed is:

Springsteen wrote Born to Run

Now you try...

PTO

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5.

```
public static void songName1 ()
{
    System.out.println("All you need is love");    // L1
    return;
}
public static void printFamousSong1 ()
{
    songName1();                                // L2
    System.out.println("was written by");          // L3
    System.out.println("Lennon and McCartney");    // L4
    return;
}
```

with a test call      `printFamousSong1 ();`

Complete the dry run tables (you may not need all rows):

| printFamousSong1() |                      |
|--------------------|----------------------|
| LINE               | printed              |
| L2                 | All you need is love |
| L3                 | was written by       |
| L4                 | Lennon and McCartney |

| songName1() |                      |
|-------------|----------------------|
| LINE        | printed              |
| L1          | All you need is love |

Write EXACTLY what is printed here

All you need is love

was written by

Lennon and Mcartney

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6.

```
public static void songName2 ()
{
    System.out.println("All you need is love");    // L1
    return;
}
public static void printFamousSong2 ()
{
    System.out.println("was written by");          // L2
    songName();                                    // L3
    System.out.println("Lennon and McCartney");    // L4
    return;
}
```

with a test call      `printFamousSong2 ();`

Complete the dry run tables (you may not need all rows):

| printfamousSong2() |                      |
|--------------------|----------------------|
| LINE               | printed              |
| L2                 | was written by       |
| L3                 | all you need is love |
| L4                 | lennon and McCartney |

| songName2() |                      |
|-------------|----------------------|
| LINE        | printed              |
| L1          | All you need is love |

Write EXACTLY what is printed here

was written by  
All you need is love  
Lennon and McCartney